

MT8002 Elements of Set Theory
7.2 Ordered Sets. Linearly Ordered Sets

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Preliminaries

Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

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Introduction

- We can use binary relations to order **some** (partial order) or **all** (linear order) the elements of a set.
- Orders are binary relations that satisfy certain properties.
- **Weak** orders and **strict** orders can be interchangeably defined.

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Binary Relations

Definition (p. 164)

Let X be a set. A set R is a **binary relation on X** if $R \subseteq X \times X$.

Notation

Let R be a binary relation on a set X .

- (a) We write $(x, y) \in R$ or xRy , if $x \in X$ is related to $y \in X$ by the relation R .
- (b) We write $(x, y) \notin R$ or $x \not R y$, if $x \in X$ is not related to $y \in X$ by the relation R .

Weak Orders

Definitions (p. 164)

A binary relation R on a set X is a **weak partial order** on the set X (or the set X is **weakly partially ordered** by the relation R), if it has the following properties:

- (a) **reflexive**: for all $x \in X$, xRx ;
- (b) **anti-symmetric**: for all $x, y \in X$, if xRy and yRx then $x \doteq y$;
- (c) **transitive**: for all $x, y, z \in X$, if xRy and yRz then xRz .

Weak Orders

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- (c) **transitive**: for all $x, y, z \in X$, if xRy and yRz then xRz .

If additionally the binary relation R has the following property:

- (d) **linear**: for all $x, y \in X$, xRy or yRx ,
- then the relation R is a **weak linear order** on the set X (or the set X is **weakly linearly ordered** by the relation R).

Weak Orders

Examples (informal, p. 164)

The usual relations $\leq$ and $\geq$ on the sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are weak linear orders on those sets.

Weak Orders

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Example

Let X be a set. The relation $\subseteq$ on $\mathcal{P}(X)$ is a weak partial order.

Weak Orders

Exercise

Let X be a set. Is the relation $\subseteq$ on $\mathcal{P}(X)$ a weak linear order?

Weak Orders

Exercise

Let X be a set. Is the relation $\subseteq$ on $\mathcal{P}(X)$ a weak linear order?

Answer

No! For example the relation $\subseteq$ on the power set of $\{a, b\}$ is no linear because $\{a\}, \{b\} \in \mathcal{P}(\{a, b\})$ but neither $\{a\} \subseteq \{b\}$ nor $\{b\} \subseteq \{a\}$.

Weak Orders

Description

The **restriction** of a weak partial order to a subset is the weak partial order obtained by discarding all comparisons involving elements outside of the subset.

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Definition (p. 164)

Let R be a weak partial order on a set X and let $A \subseteq X$. The **restriction** of the relation R to the set A , denoted $R|_{A \times A}$, is the relation defined by

$$R|_{A \times A} := R \cap A \times A \doteq \{(a, b) \in R : a, b \in A\}.$$

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$$R|_{A \times A} := R \cap A \times A \doteq \{(a, b) \in R : a, b \in A\}.$$

Theorem

Let R be a weak partial (respectively linear) order on a set X and let $A \subseteq X$. Then $R|_{A \times A}$ is a partial (respectively linear) order on the set A (Exercise 7.2).

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Strict Orders

Definitions (p. 165)

A binary relation S on a set X is a **strict partial order** on X (or the set X is **strictly partially ordered** by the relation S) if it has the following properties:

- (a) **irreflexive**: for all $x \in X$, $x \not S x$;
- (b) **transitive**: for all $x, y, z \in X$, if $x S y$ and $y S z$ then $x S z$.

Strict Orders

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A binary relation S on a set X is a **strict partial order** on X (or the set X is **strictly partially ordered** by the relation S) if it has the following properties:

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- (b) **transitive**: for all $x, y, z \in X$, if $x S y$ and $y S z$ then $x S z$.

If additionally the binary relation S has the following property:

- (c) **linear**: for all $x, y \in X$, $x S y$ or $x \doteq y$ or $y S x$,

then the relation S is a **strict linear order** on the set X (or the set X is **strictly linearly ordered** by the relation S).

Strict Orders

Examples

The usual relations $<$ and $>$ on the sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are strict linear orders on those sets.

Example

Let X be a set. The relation $\subset$ on $\mathcal{P}(X)$ is a strict partial order.

Strict Orders

Theorem (relationship between weak and strict partial orders)

(a) Let R be a weak partial order on a set X . The binary relation S defined by

$$xSy := xRy \text{ and } x \neq y$$

is a strict partial order on the set X .

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is a strict partial order on the set X .

(b) Let S be a strict partial order on a set X . The binary relation R defined by

$$xRy := xSy \text{ or } x \doteq y$$

is a weak partial order on the set X .

Strict Orders

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(a) Let R be a weak partial order on a set X . The binary relation S defined by

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is a strict partial order on the set X .

(b) Let S be a strict partial order on a set X . The binary relation R defined by

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is a weak partial order on the set X .

See Exercise 7.4.

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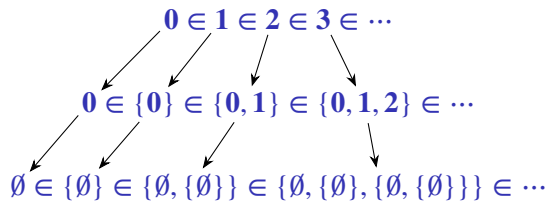
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Orders on Natural Numbers

An “extra” property of natural numbers



Orders on Natural Numbers

Theorem 3.6

The set of natural numbers $\mathbb{N}$ is strictly linearly ordered by the binary relation $\in$.

Orders on Natural Numbers

Theorem 3.6

The set of natural numbers $\mathbb{N}$ is strictly linearly ordered by the binary relation $\in$.

That is, the binary relation $\in$ on $\mathbb{N}$ has the following properties:

- (a) **irreflexive**: for all $n \in \mathbb{N}$, $n \notin n$;
- (b) **transitive**: for all $m, n, p \in \mathbb{N}$, if $m \in n$ and $n \in p$ then $m \in p$;
- (c) **linear**: for all $m, n \in \mathbb{N}$, $m \in n$ or $m \doteq n$ or $n \in m$.

Orders on Natural Numbers

Example (p. 42)

Although the binary relation $\in$ is transitive on the set $\mathbb{N}$ it does not mean that the relation is transitive on any set. For example, the relation is not transitive on the set

$$\{0, \{0\}, \{\{0\}\}\}$$

because $0 \in \{0\}$ and $\{0\} \in \{\{0\}\}$ but $0 \notin \{\{0\}\}$.

Orders on Natural Numbers

Definition (p. 41)

For all $\mathbf{m}, \mathbf{n} \in \mathbb{N}$,

$\mathbf{m} < \mathbf{n} := \mathbf{m} \in \mathbf{n}$,

$\mathbf{m} \leq \mathbf{n} := \mathbf{m} < \mathbf{n}$ or $\mathbf{m} \doteq \mathbf{n}$.

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Conventions

- “We shall often write the order relation on X as $<_X$ or $\leq_X$, depending on whether the order is *strict* or *weak*.” (p. 166)
- Sometimes a (weakly or strictly; partially or linearly) ordered set X by a relation R will we denote by the pair (X, R) .[†]
- Joining above conventions we shall write:
 - “Let $(X, \leq_X)$ be a partially (respectively linearly) ordered set” when $\leq_X$ is a *weak* partial (respectively linear) order on the set X .
 - “Let $(X, <_X)$ be a partially (respectively linearly) ordered set” when $<_X$ is a *strict* partial (respectively linear) order on the set X .

[†]Note that the textbook does not use this convention.

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Order-Isomorphism

We need a precise way to express when two ordered sets are “the same”.

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Definitions (p. 166)

Let $(X, <_X)$ and $(Y, <_Y)$ be partially ordered sets. The ordered sets are **order-isomorphic**, denoted $X \cong Y$ or $(X, <_X) \cong (Y, <_Y)$, if there is a **bijection** $f : X \longrightarrow Y$ such that for all $x_1, x_2 \in X$,

$$x_1 <_X x_2 \quad \text{if and only if} \quad f(x_1) <_Y f(x_2).$$

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The map f is an **order-isomorphism**.

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The map f is an **order-isomorphism**.

Remark

The above definition on weak partial orders is similar.

Order-Isomorphism

Exercise (informal)

Let $<$ and $>$ be the usual orders on $\mathbb{Z}$. Show that $(\mathbb{Z}, <)$ and $(\mathbb{Z}, >)$ are order-isomorphic.

Order-Isomorphism

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Let $<$ and $>$ be the usual orders on $\mathbb{Z}$. Show that $(\mathbb{Z}, <)$ and $(\mathbb{Z}, >)$ are order-isomorphic.

Solution

Let $f : \mathbb{Z} \longrightarrow \mathbb{Z}$ be the function $n \longmapsto -n$. We can show that the function f is a bijection and that it preserves the order, that is, for all $m, n \in \mathbb{Z}$,

$$m < n \quad \text{if and only if} \quad -m > -n.$$

Order-Isomorphism

Theorem

The binary relation $\cong$ is an **equivalence relation** on strictly partially ordered sets.

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That is, let $(X, <_X)$, $(Y, <_Y)$ and $(Z, <_Z)$ be partially ordered sets, then the binary relation $\cong$ satisfies the following properties (Exercise 7.6):

- (a) **reflexive**: for all X , $X \cong X$;
- (b) **symmetric**: for all X and Y , if $X \cong Y$ then $Y \cong X$;
- (c) **transitive**: for all X , Y and Z , if $X \cong Y$ and $Y \cong Z$ then $X \cong Z$.

Order-Isomorphism

Theorem

The binary relation $\cong$ is an **equivalence relation** on strictly partially ordered sets.

That is, let $(X, <_X)$, $(Y, <_Y)$ and $(Z, <_Z)$ be partially ordered sets, then the binary relation $\cong$ satisfies the following properties (Exercise 7.6):

- (a) **reflexive**: for all X , $X \cong X$;
- (b) **symmetric**: for all X and Y , if $X \cong Y$ then $Y \cong X$;
- (c) **transitive**: for all X , Y and Z , if $X \cong Y$ and $Y \cong Z$ then $X \cong Z$.

Technical remark

Note that the binary relation $\cong$ should be defined on the universe $\mathcal{U}$.

Order-Isomorphism

Definition (p. 168)

An **order-theoretic invariant** is a property of an ordered set preserved under order-isomorphism.

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Order-Embedding

We also need to define when an ordered set fits inside another while preserving its order structure entirely.

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Definition (p. 167)

Let $(X, <_X)$ and $(Y, <_Y)$ be partially ordered sets. A function $f: X \rightarrow Y$ is an **order-embedding** of X into Y (or of $(X, <_X)$ into $(Y, <_Y)$) if f is **one-one** and, for all $x_1, x_2 \in X$,

$$x_1 <_X x_2 \quad \text{if and only if} \quad f(x_1) <_Y f(x_2).$$

Order-Embedding

Exercise (informal)

Let $\mathbb{N}$ and $\mathbb{R}$ with the usual order $<$. Show an order-embedding of $(\mathbb{N}, <)$ into $(\mathbb{R}, <)$

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Let $\mathbb{N}$ and $\mathbb{R}$ with the usual order $<$. Show an order-embedding of $(\mathbb{N}, <)$ into $(\mathbb{R}, <)$

Solution

Let $f : \mathbb{N} \longrightarrow \mathbb{R}$ be the function $n \longmapsto n$. We can show that the function f is one-one and that it preserves the order, that is, for all $m, n \in \mathbb{N}$,

$$m < n \quad \text{if and only if} \quad f(m) < f(n).$$

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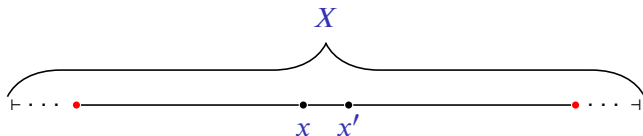
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Representation of Linearly Ordered Sets

Let $(X, <_X)$ be a linearly ordered set and let $x, x' \in X$ with $x <_x x'$. We picture x to the left of x' . The red dots mean that an element does not belong to the set X .



"Of course, we must not read too much into the picture. The picture is very suggestive of the usual real line, but for instance there might not be any elements of X between x and x' in the picture above." (p. 167)

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Maximum and Minimum Elements

Definitions (p. 168)

Let $(X, \leq_X)$ be a linearly ordered set and let $a, b \in X$.

Maximum and Minimum Elements

Definitions (p. 168)

Let $(X, \leq_X)$ be a linearly ordered set and let $a, b \in X$.

(a) The element b is the **maximum** (or **greatest**) element of X , if $x \leq_X b$ for all $x \in X$.

Maximum and Minimum Elements

Definitions (p. 168)

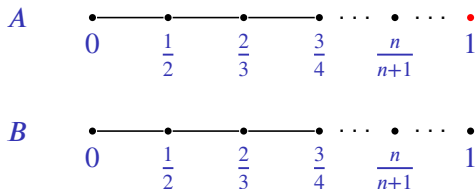
Let $(X, \leq_X)$ be a linearly ordered set and let $a, b \in X$.

- (a) The element b is the **maximum** (or **greatest**) element of X , if $x \leq_X b$ for all $x \in X$.
- (b) The element a is the **minimum** (or **least**) element of X , if $a \leq_X x$ for all $x \in X$.

Maximum and Minimum Elements

Example (informal, p. 168)

Let $A = \{1 - \frac{1}{n+1} : n \in \mathbb{N}\}$ and $B = A \cup \{1\}$, subsets of $\mathbb{Q}$, with the usual order $\leq$.



The set A has minimum element, 0 , but no maximum element.

The set B has minimum element, 0 , and maximum element, 1 .

Maximum and Minimum Elements

In the next theorem we shall see that the maximum and the minimum are order-theoretic invariants.

Maximum and Minimum Elements

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Theorem 7.1

Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be linearly ordered sets and let $f: X \longrightarrow Y$ be an order-isomorphism from X to Y . Then

- (a) if X has a maximum element then so does Y ;
- (b) if X has a minimum element then so does Y .

Maximum and Minimum Elements

Example (informal)

Let $A = \{1 - \frac{1}{n+1} : n \in \mathbb{N}\}$ and $B = A \cup \{1\}$, subsets of $\mathbb{Q}$, with the usual order $\leq$.

By Theorem 7.1 the ordered sets $(A, \leq)$ and $(B, \leq)$ are not order-isomorphic because the set B has maximum element, but the set A does not.

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Limit Points

Definition (p. 169)

Let $(X, <_X)$ be a linearly ordered set and let $c \in X$. The element c is a **limit point** of X if

- (a) there exists $x \in X$ such that $x <_X c$ and
- (b) for all $x \in X$ with $x <_X c$, there exists $y \in X$ such that $x <_X y <_X c$.

Limit Points

In the next theorem we shall see that limit points are order-theoretic invariants.

Limit Points

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Theorem 7.2

Let $(X, <_X)$ and $(Y, <_Y)$ be linearly ordered sets, let $f : X \longrightarrow Y$ be an order-isomorphism from X to Y and let $c \in X$. If c is a limit point of X then $f(c)$ is a limit point of Y (Exercise 7.11).

Limit Points

Exercise (informal, p. 170)

Let A and D be subsets of $\mathbb{Q}$, with the usual order $<$, defined by

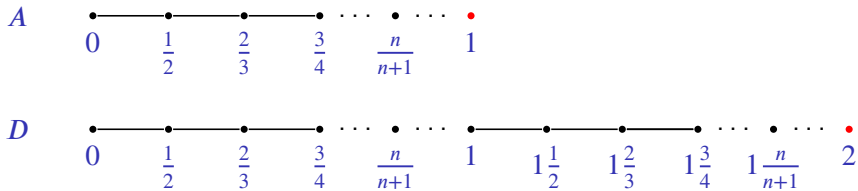
$$A = \left\{ 1 - \frac{1}{n+1} : n \in \mathbb{N} \right\},$$
$$D = \left\{ 1 - \frac{1}{n+1} : n \in \mathbb{N} \right\} \cup \left\{ 2 - \frac{1}{n+1} : n \in \mathbb{N} \right\}.$$

Are $(A, <)$ and $(D, <)$ order-isomorphic?

(continued on next slide)

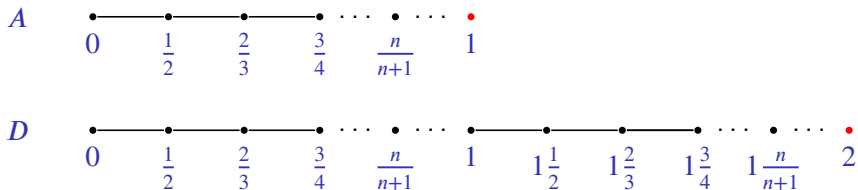
Limit Points

Solution



Limit Points

Solution



- (a) The element $0 = 1 - \frac{1}{0+1}$ is the **minimum** (so 0 is not a limit point) of the sets A and D .
- (b) The set A **has not limit points**.
- (c) The element $1 = 2 - \frac{1}{0+1}$ is a **limit point** of set D .

Therefore, since $(B, <)$ has one limit point but $(A, <)$ does not have, the ordered sets are not order-isomorphic by Theorem 7.2.

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Successors and Predecessors

Definition (p. 170)

Let $(X, <_X)$ be a linearly ordered set and let $x, d \in X$. The element d is the **successor** of x , denoted x^+ , and x is the **predecessor** of d , if

- (a) $x <_X d$ and
- (b) for all $y \in X$, if $x <_X y$ then $d \doteq y$ or $d <_X y$.

Successors and Predecessors

Definition (p. 170)

Let $(X, <_X)$ be a linearly ordered set and let $x, d \in X$. The element d is the **successor** of x , denoted x^+ , and x is the **predecessor** of d , if

- (a) $x <_X d$ and
- (b) for all $y \in X$, if $x <_X y$ then $d \doteq y$ or $d <_X y$.

That is, d is the successor of x (and x is the predecessor of d) if d is the first element of X greater than x .

Successors and Predecessors

Questions

In a strictly linearly ordered set

(a) can a limit point also be a successor?

Successors and Predecessors

Questions

In a strictly linearly ordered set

- (a) can a limit point also be a successor?
- (b) can a successor also be a limit point?

Successors and Predecessors

Examples (informal)

(a) In $\mathbb{R}$ with the usual order $<$, every real number is a limit point, so $\mathbb{R}$ has no successors.

Successors and Predecessors

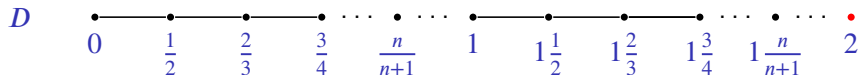
Examples (informal)

- (a) In $\mathbb{R}$ with the usual order $<$, every real number is a limit point, so $\mathbb{R}$ has no successors.
- (b) In $\mathbb{N}$ with the usual order $<$, every natural number different from zero is a successor, so $\mathbb{N}$ has no limit points.

Successors and Predecessors

Example

Let $D = \{1 - \frac{1}{n+1} : n \in \mathbb{N}\} \cup \{2 - \frac{1}{n+1} : n \in \mathbb{N}\}$ be a subset of $\mathbb{Q}$, with the usual order $<$.



- (a) The element $0 = 1 - \frac{1}{0+1}$ is the **minimum** (0 is neither a successor nor a limit point).
- (b) Every element $1 - \frac{1}{n+1}$, with $n > 0$, is the **successor** of the element $1 - \frac{1}{n}$.
- (c) The element $1 = 2 - \frac{1}{0+1}$ is a **limit point**.
- (d) Note that for all $x \in D$, with $x < 1$, x has a **successor** x^+ and the set $\{x \in D : x < 1\}$ is **infinite** (see Exercise 7.14).
- (e) Every element $2 - \frac{1}{n+1}$, with $n > 0$, is the **successor** of the element $2 - \frac{1}{n}$.

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Definition (p. 188)

Let $(X, <_X)$ be a linearly ordered set. The relation $<_X$ is a **well-order** (or the set X is **well-ordered** by the relation $<_x$) if every non-empty subset A of X has a least element, i.e., there is some element $x \in A$ such that $x \leq_X a$, for all $a \in A$.

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Let $(X, <_X)$ be a linearly ordered set. The relation $<_X$ is a **well-order** (or the set X is **well-ordered** by the relation $<_x$) if every non-empty subset A of X has a least element, i.e., there is some element $x \in A$ such that $x \leq_X a$, for all $a \in A$.

Theorem 3.9

The set $\mathbb{N}$ is well-ordered by the relation $\in$.

Finite Strictly Linearly Ordered Sets

The following theorem will state that every strictly linearly ordered **finite** set is order-isomorphic to some natural number **n** ordered by **$\in$** .

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Theorem

Let **$n \in \mathbb{N}$** and let **$(X, <_X)$** be a linearly ordered finite set with **n** elements. Then **$(X, <_X) \cong (n, \in)$** (Exercise 7.15(a)).

Outline

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Successors and Predecessors

Finite Strictly Linearly Ordered Sets

References

References

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).